Numerical simulations of the aerodynamics of a speedway motorcycle: Supplementary Material

Ivan Pujol Vidal[a,*], Sebastian Obwoya Gudmand-Høyer[a], Yvet Maathuis[a,b], Andreas Top Adler[b], Robert Flemming Mikkelsen[c] and Jens Honoré Walther[a,**]

[a]Department of Civil and Mechanical Engineering, Technical University of Denmark, Nils Koppels Allé 403, Kongens Lyngby, 2800, Denmark
[b]Team Danmark, Idrættens Hus Brøndby Stadion 20, Brøndby, 2605, Denmark
[c]Department of Wind and Energy Systems, Technical University of Denmark, Nils Koppels Allé 403, Kongens Lyngby, 2800, Denmark

1. Convergence studies

To ensure numerical accuracy and reliability, a series of convergence studies are performed, assessing the sensitivity of the results to mesh resolution and temporal discretisation. These tests establish independence of the solution from numerical settings while balancing computational efficiency and accuracy.

1.1. Mesh convergence

The mesh convergence analysis is divided into two parts: base size sensitivity and prism layer refinement. These studies ensure that the drag force predictions are independent of the grid resolution and that near-wall effects are properly captured.

1.1.1. Base size convergence

The base size convergence study assessed the effect of global mesh refinement on drag predictions. Six meshes are generated with base sizes ranging from 80 to 130 mm, while keeping all other parameters fixed. The number of prism layers is set to six with a total thickness of 2 mm, and the time step is fixed at 1×10^{-4} s to ensure that the finest mesh maintained acceptable CFL values.

All simulations are run for 0.5 s, approximately 7.25 times the dimensionless parameter tU/L, where t is time, U is the free stream velocity, and L is the length of the geometry. As shown in Figure 1, at that point, drag area is fully stabilized. Values before 0.1 s (approximately 1.45 times tU/L) are discarded to calculate the mean results.

The average drag forces for all base sizes are summarised in Figure 2. A fitted regression line is used to extrapolate results towards an idealised case of zero base size, predicting a drag area of 0.51.

The results indicate that coarse meshes (120–130 mm) deviate by approximately 2.5% from the extrapolated ideal case, which is deemed too high. In contrast, the 110 mm mesh shows a deviation of only 1.9%, and the difference compared to the finest tested mesh (80 mm) is just 0.25%. These variations are considered negligible, confirming that the 110 mm base size provides a robust compromise between accuracy and computational cost. This resolution is therefore selected for the final mesh.

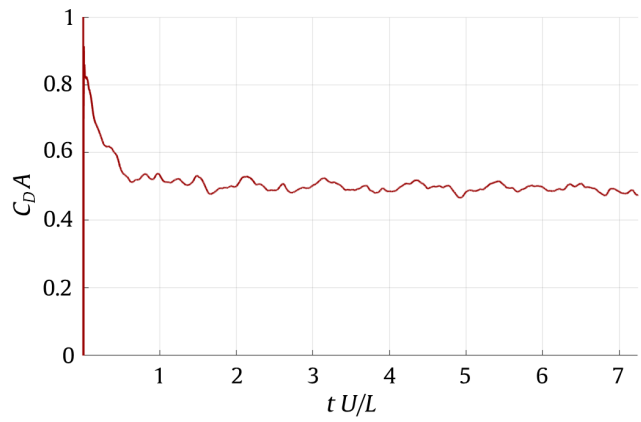

Figure 1: Drag area values over time

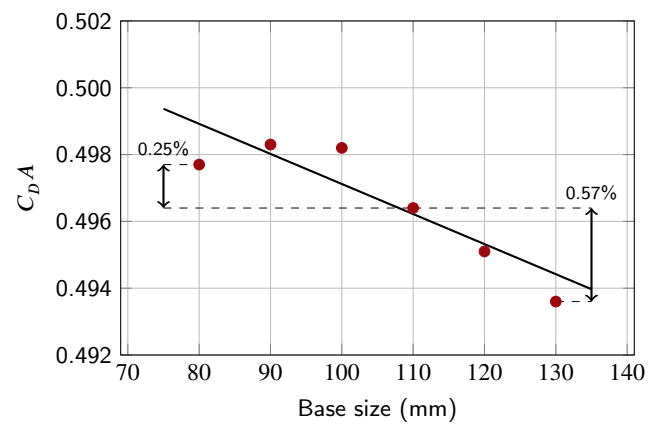

Figure 2: Drag area vs Base size. Percentage errors comparing the selected base size with the finest and coarsest mesh cases. Fitted function: $f(x) = -0.00009x + 0.506$

1.1.2. Number of prism layers

Once the base size is established, the number of prism layers is varied to assess the resolution of the near-wall region. Five simulations are performed with prism layer number between four and eight, maintaining a constant total thickness of 2 mm. The wall y^+ values are monitored in all cases, as this parameter dictates whether the near-wall

*Corresponding author
**Principal corresponding author
ivanpujol0407@gmail.com (I. Pujol Vidal); s19502@dtu.dk (S.O. Gudmand-Høyer); yvma@dtu.dk (Y. Maathuis); atad@teamdanmark.dk (A. Top Adler); rfmi@dtu.dk (R.F. Mikkelsen); jhwa@dtu.dk (J.H. Walther)
ORCID(s): 009-0001-2159-9388 (Y. Maathuis); 0000-0002-4095-1228 (R.F. Mikkelsen); 0000-0001-8100-9178 (J.H. Walther)

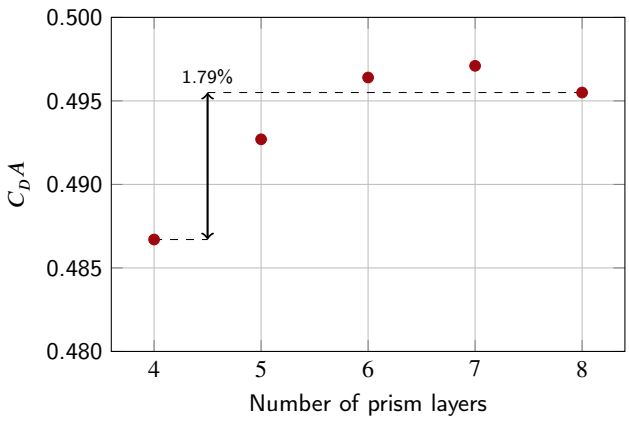

Figure 3: Average drag area vs Number of prism layers. Percentage error comparing highest and lowest number of prism layers

cells lie within the viscous sublayer and thus whether the turbulence model can accurately capture boundary layer effects. The wall y^+ value is a dimensionless parameter that represents the distance from the first grid cell to the surface wall. It is calculated using the Equation 1:

$$y^+ = \frac{u_\tau y}{\nu}, \tag{1}$$

where y is the distance to the first cell centre, ν is the kinematic viscosity, and $u_\tau = \sqrt{\tau_w/\rho}$ is the friction velocity based on the wall shear stress τ_w and fluid density ρ. Maintaining y^+ within the appropriate range is essential for turbulence modelling. For low-Reynolds number applications and for models such as the k–ω SST model used in this study, best practice dictates keeping most values within the viscous sublayer ($y^+ < 5$), ideally around unity [1, 3].

Figure 3 presents the resulting drag forces for each configuration. The variation in drag force across the tested cases remained below 2%. Such variations are considered negligible, indicating that all tested configurations are numerically valid.

Despite the small drag variation, wall y^+ distributions revealed clear differences in near-wall resolution. With 4 or 5 prism layers, a significant fraction of cells exhibited y^+ values above 5 as seen in the histograms in Figures 4 and 5, placing them in the buffer layer. While still numerically acceptable, this configuration reduces the ability of the turbulence model to resolve boundary layer effects accurately. Increasing the prism count to 6 and 7 layers improved the distribution, but some cells remained above the recommended threshold, as shown in Figures 6 and 7. Only the 8-layer case ensured that all the wall-adjacent cells are located within the viscous sublayer ($y^+ < 5$), as shown in the histogram in Figure 8. Importantly, increasing the prism layers from 7 to 8 results in only a modest rise in total cell count, from 19.8 to 21.2 million, corresponding to a minor increase in computational cost.

In summary, the mesh convergence study confirmed that a base size of 110 mm combined with 8 prism layers offers a reliable compromise between computational efficiency and accuracy, ensuring mesh-independent results for the aerodynamic analysis.

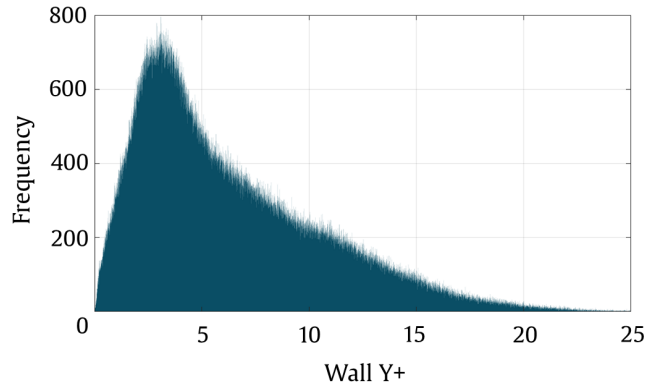

Figure 4: Wall Y+ histogram for 4 prism layers

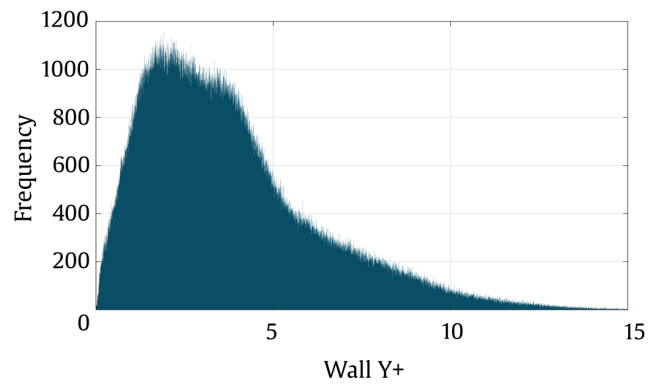

Figure 5: Wall Y+ histogram for 5 prism layers

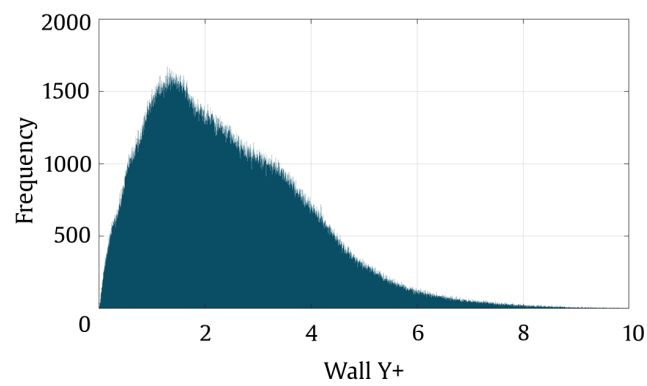

Figure 6: Wall Y+ histogram for 6 prism layers

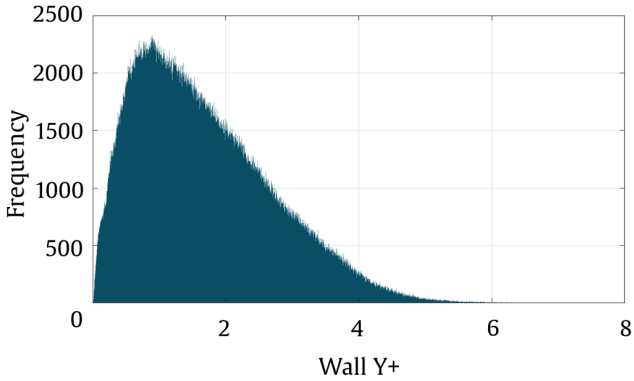

Figure 7: Wall Y+ histogram for 7 prism layers

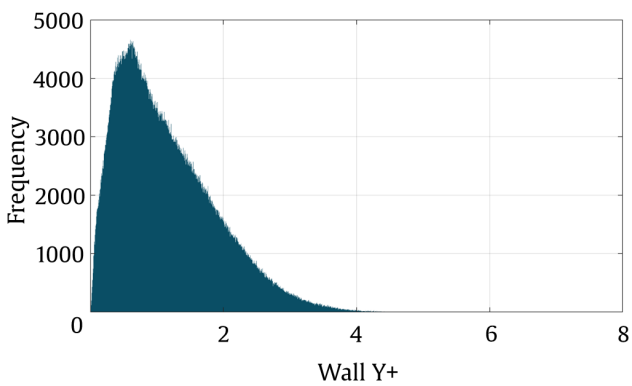

Figure 8: Wall Y+ histogram for 8 prism layers

1.2. Time step convergence

The final convergence study addressed temporal discretisation, with the objective of identifying a time step that ensures both numerical stability and temporal accuracy while maintaining computational efficiency. Time step size has a direct impact on the ability of the solver to capture transient aerodynamic effects, as excessively large steps can lead to numerical instability or under-resolved flow phenomena, whereas excessively small steps increase computational costs without significantly improving accuracy.

The Courant-Friedrichs-Lewy (CFL) condition provides a fundamental criterion for assessing time step suitability. The CFL number is a dimensionless parameter that represents the ratio of the physical distance travelled by fluid particles during one time step to the size of the computational grid cell and is defined following Equation 2:

$$CFL = \frac{u\Delta t}{\Delta x}, \tag{2}$$

where u is the local flow velocity, Δt the time step size, and Δx the grid size. To maintain accuracy, the CFL number is generally required to remain below unity [1, 2]. In this study, the CFL number is continuously monitored to ensure compliance with this condition.

Four different time steps are tested: 5×10^{-5} s, 1×10^{-4} s, 2×10^{-4} s, and 4×10^{-4} s. Each simulation is performed with the previously validated mesh (base size 110 mm, 8 prism layers) and run for 0.5 s of physical time. The average drag area is extracted from each case after discarding the initial 0.1 s of startup transients to ensure that only steady-state conditions are considered in the comparison.

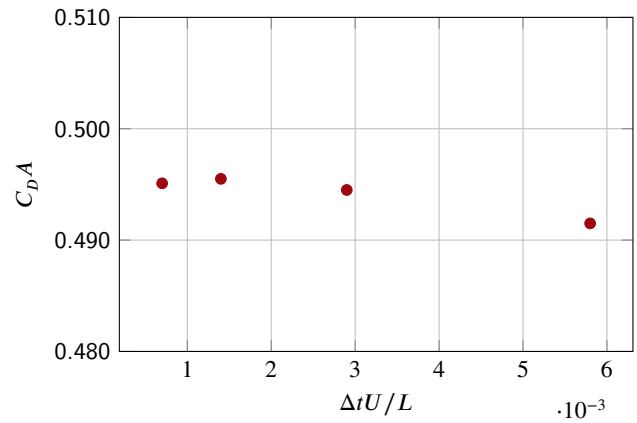

Figure 9: Drag area vs Non-dimensional time step $\Delta tU/L$

Figure 9 shows the results of the time step convergence study. The average drag area varies by less than 1% across the entire range of tested time steps, confirming that all cases are valid. The absence of a systematic and clear trend with decreasing time step further indicates that the solution is not sensitive to temporal discretisation within this range.

Although all tested configurations yield valid results, the CFL condition remains an important constraint for selecting the most appropriate time step. To ensure $CFL < 1$, contours on the central plane of the motorcycle are analysed for all tested time steps in Figures 10, 13, 12, and 11. At $\Delta t = 4 \times 10^{-4}$ s, large regions of the domain exceed $CFL = 1$, as illustrated in Figure 13, discarding this time step from consideration. Reducing the time step to 5×10^{-5} s ensures $CFL < 1$ throughout the domain, as shown in Figure 10. However, as shown in Figure 9, this refinement yields no significant change in the results. At $\Delta t = 2 \times 10^{-4}$ s,illustrated in Figure 12, CFL values remain well below unity across most of the domain, with elevated values appearing only in highly localised regions around the front wheel, lower fairing, and rider's helmet. These areas correspond to regions of high local velocity and confirm that the elevated CFL values result from flow acceleration rather than grid inconsistencies, since mesh resolution is comparable to the rest of the geometry.

The comparison across all tested time steps indicates that further reduction in Δt does not improve drag predictions, and the small regions of elevated CFL values at $\Delta t = 2 \times 10^{-4}$ s can be considered negligible. Accordingly, a time step of 2×10^{-4} s is selected for all subsequent simulations, as it satisfies the CFL condition, resolves the relevant unsteady aerodynamic effects with sufficient accuracy, and provides a balanced compromise between fidelity and computational cost.

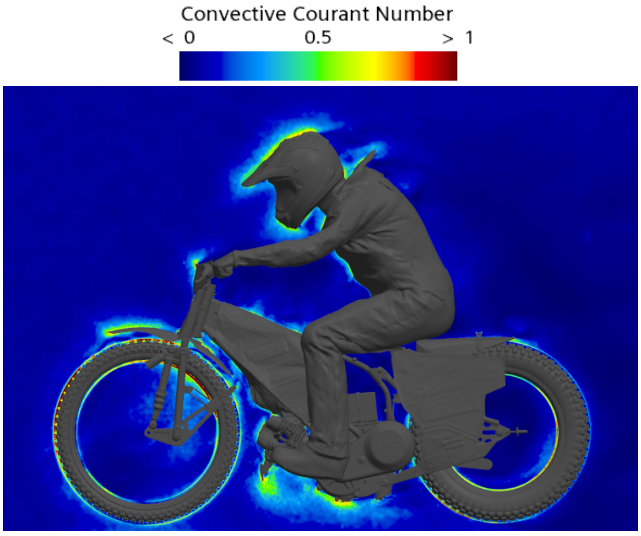

Figure 10: Courant number contour, 5×10^{-5} s time step

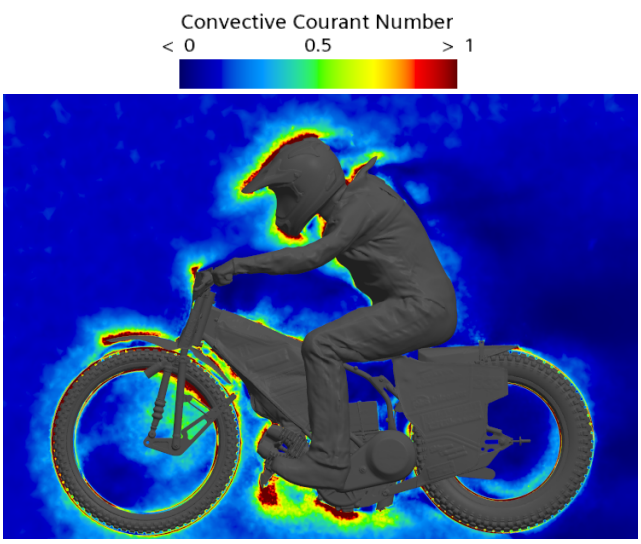

Figure 11: Courant number contour, 1×10^{-4} s time step

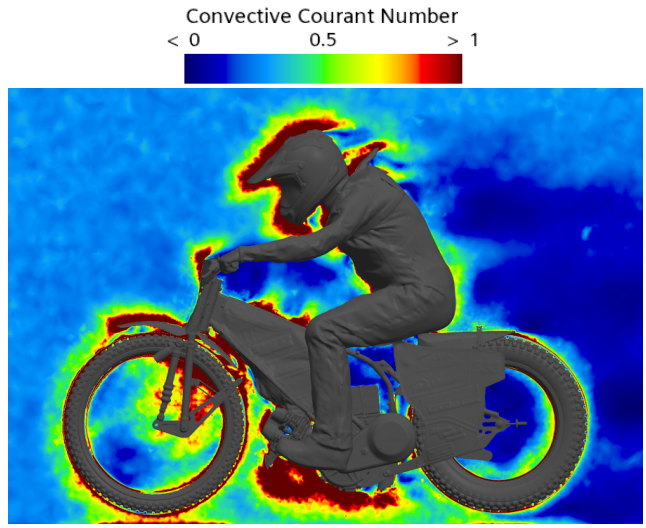

Figure 12: Courant number contour, 2×10^{-4} s time step

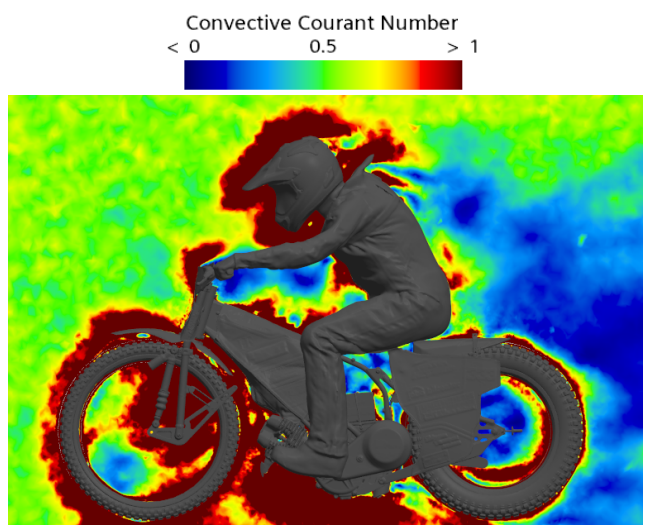

Figure 13: Courant number contour, 4×10^{-4} s time step

References

[1] van Druenen, T., Blocken, B., 2024. CFD simulations of cyclist aerodynamics: Impact of computational parameters. J. Wind Eng. Ind. Aerodyn. 249, 104714.

[2] Ferziger, J.H., Perić, M., Street, R.L., 2020. Computational Methods for Fluid Dynamics. 4 ed., Springer-Verlag.

[3] Salim, M., Cheah, S., 2009. Wall y+ strategy for dealing with wall-bounded turbulent flows, in: In Proceedings of the International MultiConference of Engineers and Computer Scientists,, Hong Kong, China. p. 2175.